

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data Mining
(Subclass C)

Date: 09 Dec, 2024

Time: 6.30pm – 8.30pm

University No:

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This is a closed book examination.

Answer ALL questions.

Write your answers on the exam paper in the corresponding space after the question.

Write your university number on every page.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of calculator: _____

Question 1 (30 marks)

(a) Please explain the concept of overfitting. (5 marks)

(b) Please explain how edge connections between nodes in graph-structured data can be effectively represented in graph machine learning. (5 marks)

(c) Please describe how graph embeddings differ from word embeddings. (5 marks)

(d) Please identify three fundamental graph learning tasks and their corresponding real-world applications. (5 marks)

(e) Identify and explain the critical stages in your search engine pipeline, including their functionalities. Then, propose three crucial metrics for evaluating search results quality. (5 marks).

(f) Please explain the key differences between training from scratch, model fine-tuning, and prompting in Large Language Models (LLMs), and their suitable application scenarios. (5 marks)

Question 2 (40 marks)

(a) An e-commerce platform has developed a customer loyalty prediction model that generates a binary classification (Yes/No) based on three key metrics. The first metric is the user's consumption activity rating, categorized as A, B, or C. The second metric measures weekly online shopping duration, divided into two groups: more than 70 hours or less than 70 hours. The third metric evaluates the annual customer satisfaction score, segmented into three ranges: below 60, between 60-80, and above 80. These metrics collectively determine whether a user will be classified as a loyal customer of the platform. (20 marks).

ID	Weekly Online Shopping Hours	Annual Satisfaction Score	User Consumption Activity Rating	Voted
1	More than 70 hours	60-80	A	T
2	Less than 70 hours	60-80	B	F
3	More than 70 hours	60-80	C	F
4	More than 70 hours	60-80	A	T
5	Less than 70 hours	<60	C	F
6	More than 70 hours	>80	B	T
7	Less than 70 hours	60-80	A	F
8	More than 70 hours	60-80	B	T
9	Less than 70 hours	60-80	C	F
10	Less than 70 hours	>80	A	T
11	More than 70 hours	<60	B	F
12	Less than 70 hours	60-80	C	F
13	Less than 70 hours	<60	B	F
14	More than 70 hours	>80	C	T

Q1: Calculate the entropy $H(\text{Voted})$ for the loyalty classification.

Q2: Determine the conditional entropy $H(\text{Voted} \mid \text{Weekly Online Shopping Hours})$.

Q3: Calculate the conditional entropy $H(\text{Voted} \mid \text{Annual Satisfaction Score})$.

Q4: Determine the conditional entropy $H(\text{Voted} \mid \text{User Consumption Activity Rating})$.

Q5: Construct a complete decision tree diagram for this dataset. Detailed calculations are not required.

(b) K-means is a fundamental clustering algorithm widely adopted in data analysis. Its core purpose is to partition data points into k distinct clusters, where each point belongs to the cluster with the nearest mean, effectively minimizing within-cluster variance. (20 marks).

Q1. Suppose we have six two-dimensional data points: $P_1(1,1)$, $P_2(2,1)$, $P_3(4,3)$, $P_4(5,4)$, $P_5(3,3)$, and $P_6(6,5)$. Apply the k-means algorithm with $k=2$, using $P_5(3,3)$ and $P_6(6,5)$ as initial centroids. Determine the final two clusters by showing your step-by-step process, including point assignments and centroid recalculations at each iteration, until the algorithm converges to its final clustering solution.

Q2. Given a set of six points in a 2D plane [$P_1(1,1)$, $P_2(2,1)$, $P_3(4,3)$, $P_4(5,4)$, $P_5(3,3)$, $P_6(6,5)$], implement a special clustering procedure where $P_1(1,1)$ serves as the pre-designated first cluster center. The task requires selecting a second center from the remaining points (P_2 - P_6) based on the maximum-distance criterion from P_1 , ensuring optimal conditions for subsequent clustering operations. This selection process aims to establish the most effective spatial distribution of cluster centers.

(i) Select an optimal second cluster center from points P_2 - P_6 by systematically calculating the Euclidean distance between each candidate point and the first center P_1 . Compare all distance values to identify and select the point that maximizes its separation from P_1 . (Hint: Obtain the likelihood of each point being selected by calculating distance between each remaining point and P_1 .)

(ii) Analyze whether this likelihood-based selection method would always yield the same point as the second cluster center, providing clear reasoning for your answer.

Question 3 (30 marks)

(a) You are a data scientist at a large e-commerce platform tasked with improving its recommendation system. Given millions of users and items with rich interaction data (purchases, views, ratings), design a recommendation system using Graph Neural Networks (GNNs) to model user-item interaction patterns, generate high-quality user embeddings, and make personalized recommendations. In your solution, address the following aspects: (1) the construction of user-item interaction graphs, (2) GNN architecture design for capturing both local and global interaction patterns, (3) advantages of GNNs over traditional embedding-based methods. (15 marks).

(b) As a technical scientist at a financial services company, design a comprehensive RAG (Retrieval-Augmented Generation) system that leverages Large Language Models while maintaining data privacy, as direct fine-tuning with sensitive financial data is not permitted. Detail your technical solution including the RAG pipeline architecture, document processing approach, embedding strategies, retrieval mechanisms, and prompt engineering methods. (15 marks).

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